

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester IV Examination, 2013-14

Paper: BSC-04A — Elements of Geology - II (Ancillary)

Time: 3 Hours

Maximum Marks: 70

Ref. No.: BSC04A Ancillary

*Instructions: Answer **five** questions in all, selecting at least **two** each from Sections A and B, and Section-C which is compulsory. The figures in the right-hand margin indicate marks.*

SECTION — A

Question 1. Explain how Palaeontology supports the phenomenon of “Organic Evolution”.
[14]

Question 2. Discuss petrification and its kinds. What change(s) occur(s) with respect to the original organism?
[14]

Question 3. Write explanatory notes on **any four** of the following: $[4 \times 3\frac{1}{2} = 14]$

- (a) Subdisciplines of Palaeontology
- (b) Distinction between a true shell and a cast
- (c) The process of carbonization. What deformation does the fossil undergo with respect to the actual organism?
- (d) What are the TWO major requirements for fossilization?
- (e) Basic differences between ‘Coprolite’ and ‘Gastrolith’.

SECTION — B

Question 4. Discuss the physical divisions of India with a neat diagram. [14]

Question 5. What is the Geological Time Scale? Write a note on how it was developed and its various uses. [14]

Question 6. Define mass extinction. How many mass extinctions are generally believed to have occurred on the Earth and during which geological time periods? Add a note on the possible causes of mass extinction. [14]

SECTION — C

Question 7. Answer the following questions: [14]

- (a) Explain the etymology of Palaeontology.

- (b) What is the oldest trace fossil?
- (c) Which of the following is/are 'true' Body Fossil(s)?
 - (i) Mesozoic dinosaur footprints
 - (ii) Chemically preserved 'Mummies' of Pharaohs
 - (iii) Oligocene insects preserved in amber
 - (iv) Pleistocene fossil 'Mammoths' of Tundra
- (d) Define a "Steinkern".
- (e) Name a 'metallic' mineral deposit discovered through the study of fossils.
- (f) Tarai-Bhabhar
- (g) The principle of superposition
- (h) Uniformitarianism
- (i) Name the important Upper Gondwana plant fossils.
- (j) Barren Measures